

第八讲：图像分割（2）



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2026-3-26

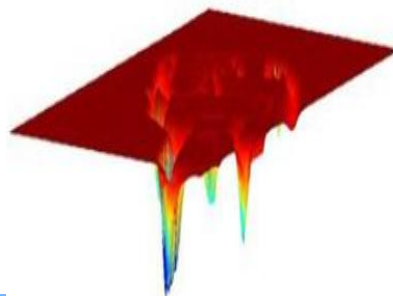
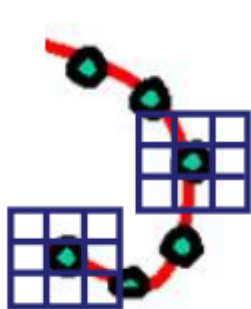
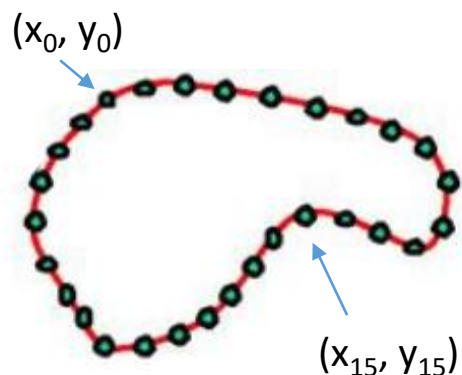


知识回顾: Snake model 蛇形模型

Snake model represents contours as a list of vertices

Internal energy: encourage smoothness and low curvature

External energy: drive the contour to edges in the image



内部能量 外部能量

$$E_{total} = E_{internal} + E_{external}$$

$$E_{internal}(v(s)) = \underbrace{\alpha \left| \frac{dv}{ds} \right|^2}_{\text{Tension, Elasticity}} + \underbrace{\beta \left| \frac{d^2v}{d^2s} \right|^2}_{\text{Stiffness, Curvature}}$$

弹性约束 曲率约束

$$E_{external} = -(G_x(I)^2 + G_y(I)^2)$$

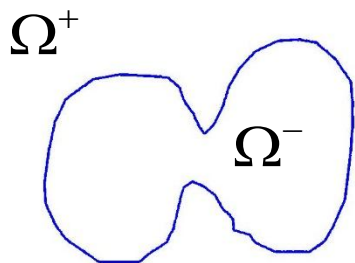
知识回顾：Level Set 水平集模型

Level set method represents the curve in the form of an **implicit surface**

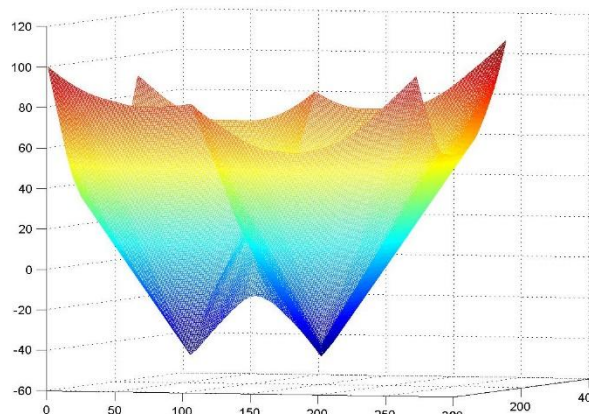
将曲线转换成曲面：带符号距离变换

$$\phi_{t+1} = \phi_t + \lambda \frac{\partial \phi}{\partial t} \quad \frac{\partial \phi}{\partial t} = F |\nabla \phi|$$

Contour C



Signed distance function



速度函数：可根据梯度或曲率来定义

$$\frac{\partial \phi}{\partial t} = c |\nabla \phi|$$

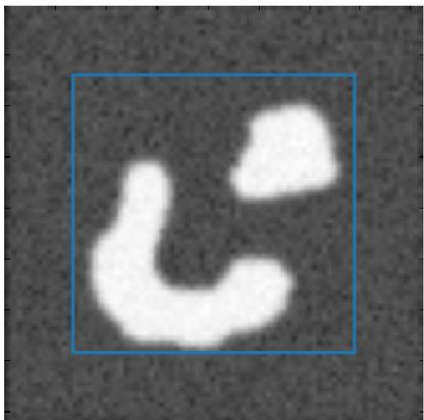
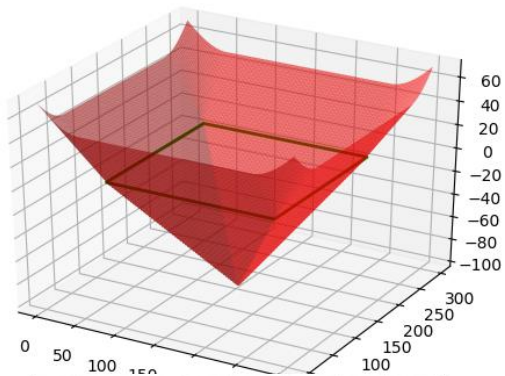
$$\frac{\partial \phi}{\partial t} = \text{div} \left(\frac{\nabla \phi}{|\nabla \phi|} \right) |\nabla \phi|$$

$$\frac{\partial \phi}{\partial t} = g(I) \|\nabla \phi\| \text{div} \left(\frac{\nabla \phi}{\|\nabla \phi\|} \right) + g(I) \|\nabla \phi\| v + \nabla g(I) \cdot \nabla \phi$$

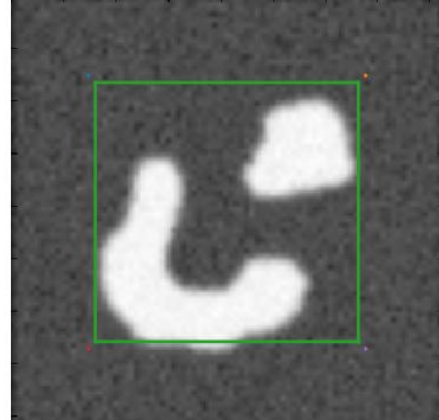
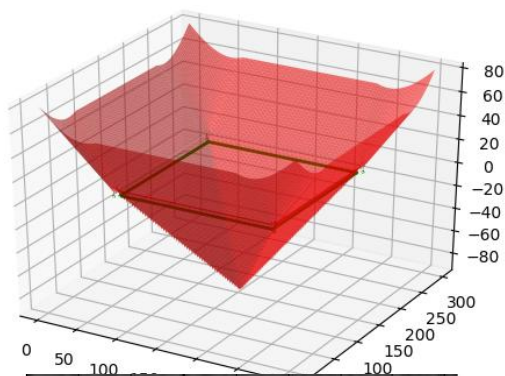
Level set 水平集

- Constant speed $\frac{\partial \phi}{\partial t} = c |\nabla \phi|$

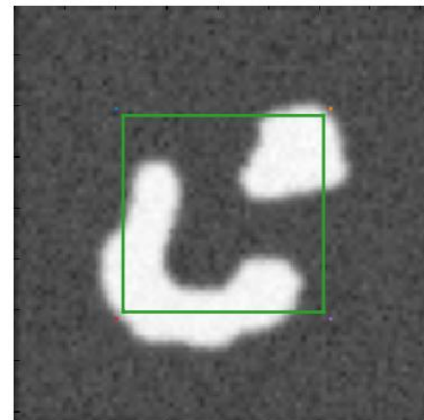
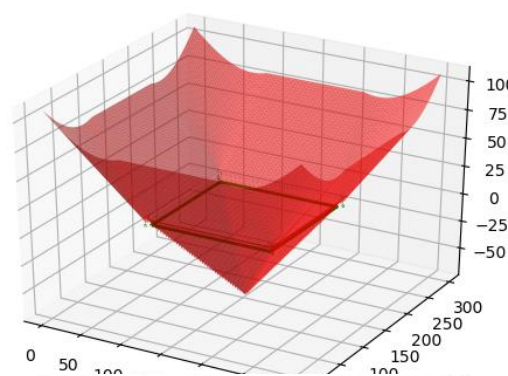
$$\phi_{t+1} = \phi_t + \lambda \frac{\partial \phi}{\partial t}$$



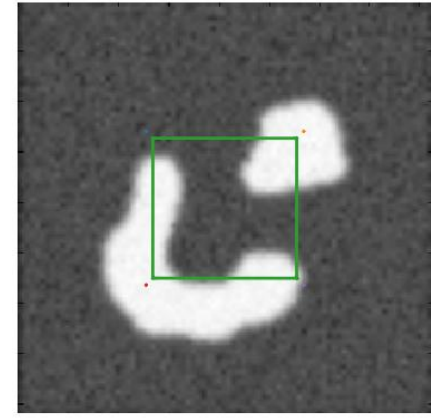
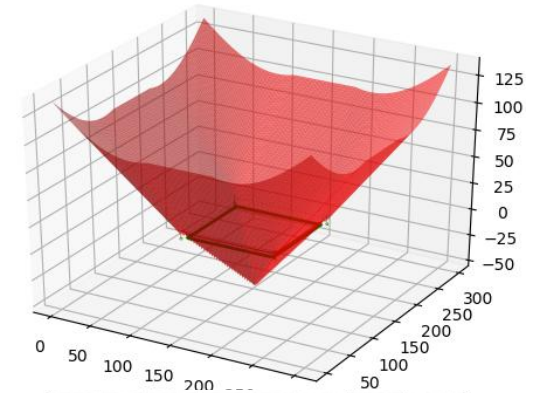
T = 0



T = 10



T = 30



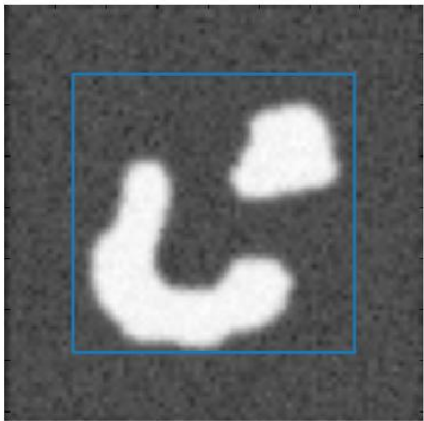
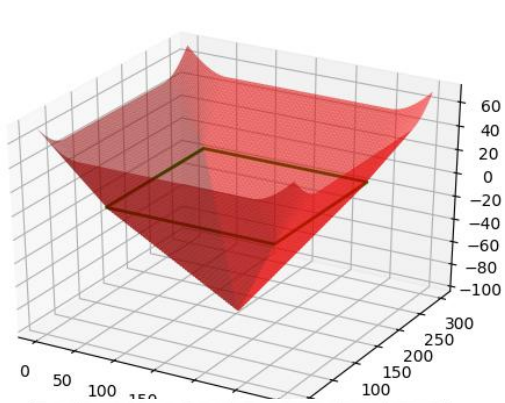
T = 50

Level set 水平集

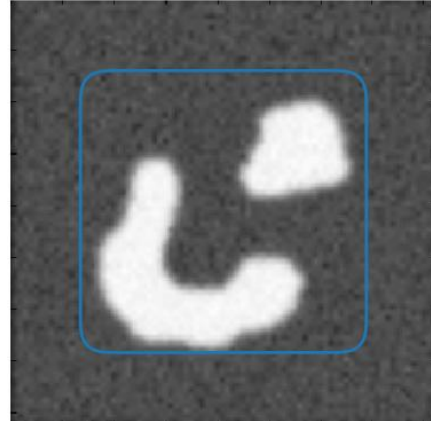
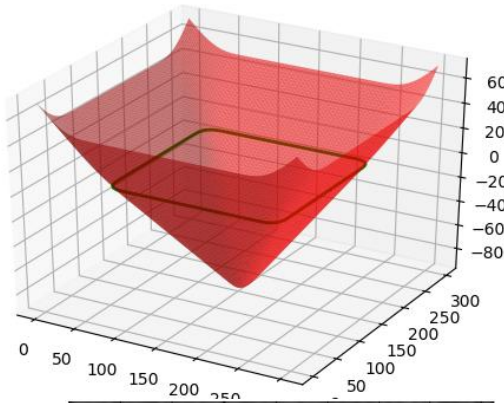
- Mean curvature flow

$$\frac{\partial \phi}{\partial t} = \operatorname{div} \left(\frac{\nabla \phi}{|\nabla \phi|} \right) |\nabla \phi|$$

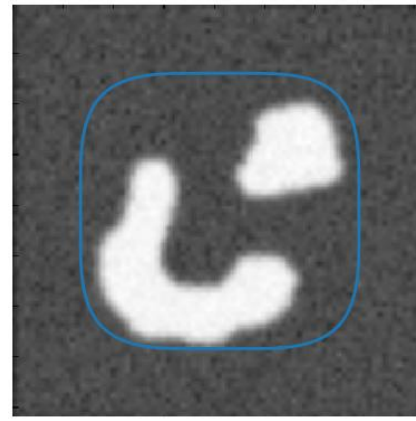
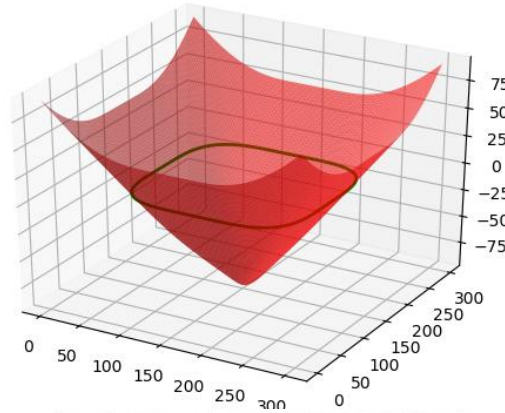
$$\phi_{t+1} = \phi_t + \lambda \frac{\partial \phi}{\partial t}$$



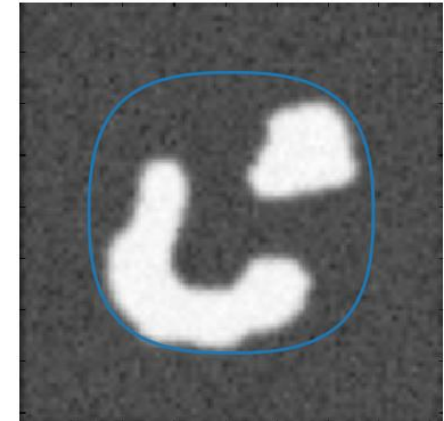
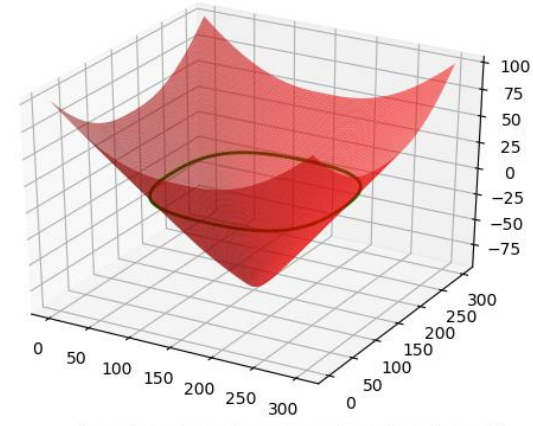
$T = 0$



$T = 50$



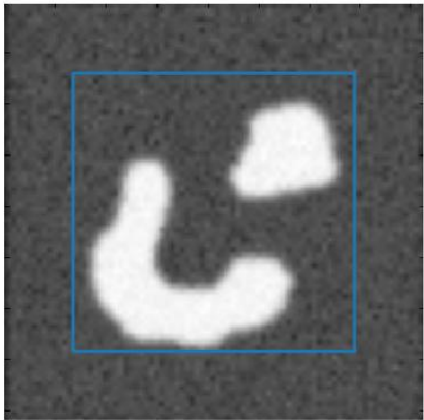
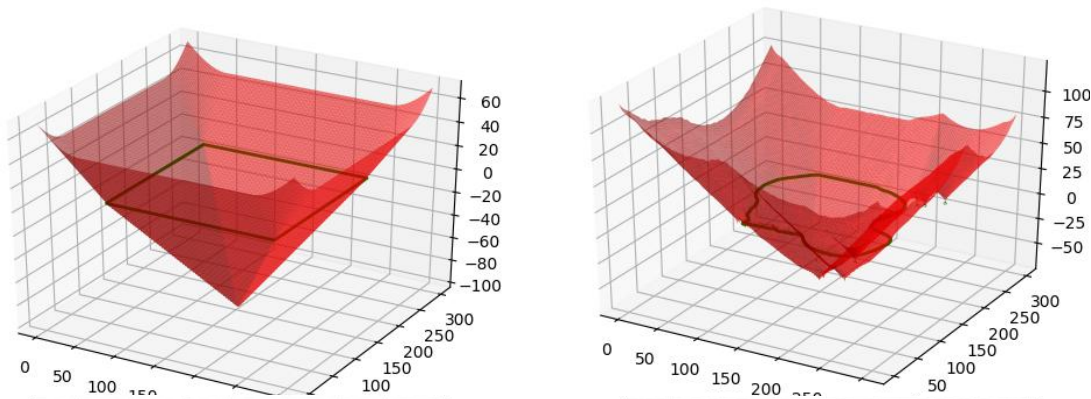
$T = 200$



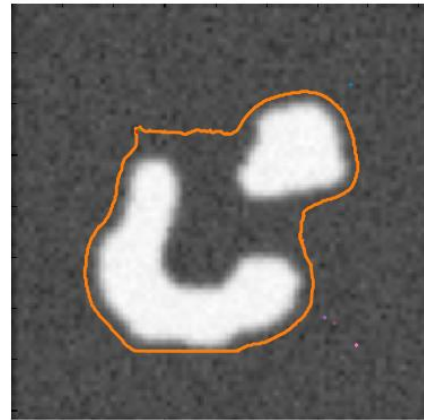
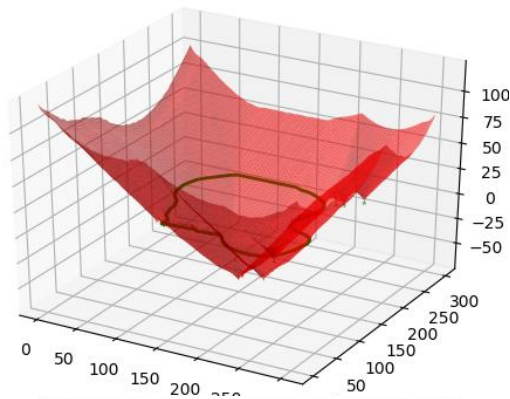
$T = 1000$

Level set 水平集

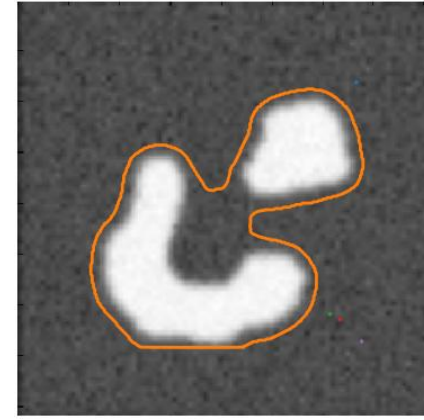
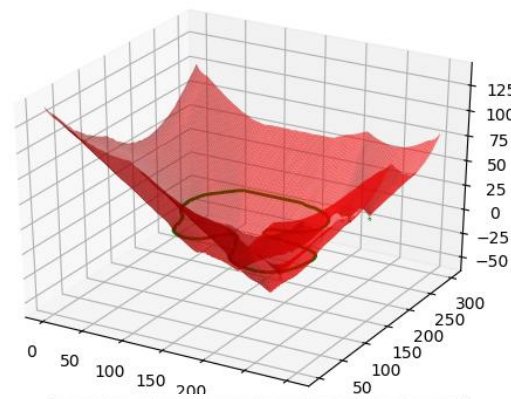
- Define the speed based on gradient $\frac{\partial \phi}{\partial t} = F|\nabla \phi|$ Let $F = g(I) = \frac{1}{1 + \|\nabla I\|^2}$



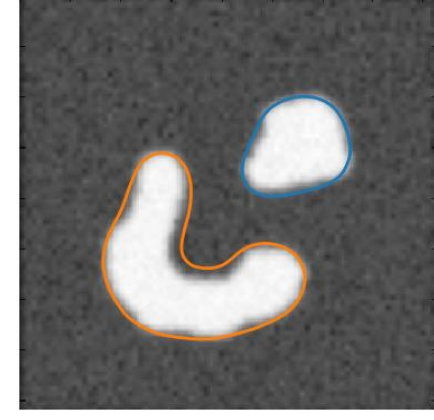
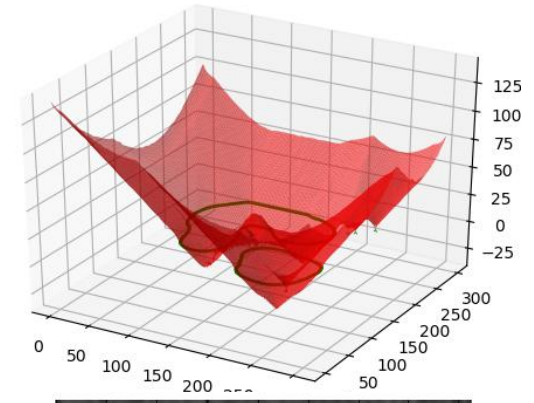
$T = 0$



$T = 50$



$T = 100$



$T = 200$

Region-based Level set 基于区域的水平集



Active contours without edges

TF Chan, LA Vese - IEEE Transactions on image processing, 2001 - ieeexplore.ieee.org

... edges too. In this paper, we propose a different active contour model, without a stopping edge-... In this way, we obtain a model which can detect contours both with or without gradient, for ...

☆ Save Cite Cited by 13614 Related articles All 50 versions

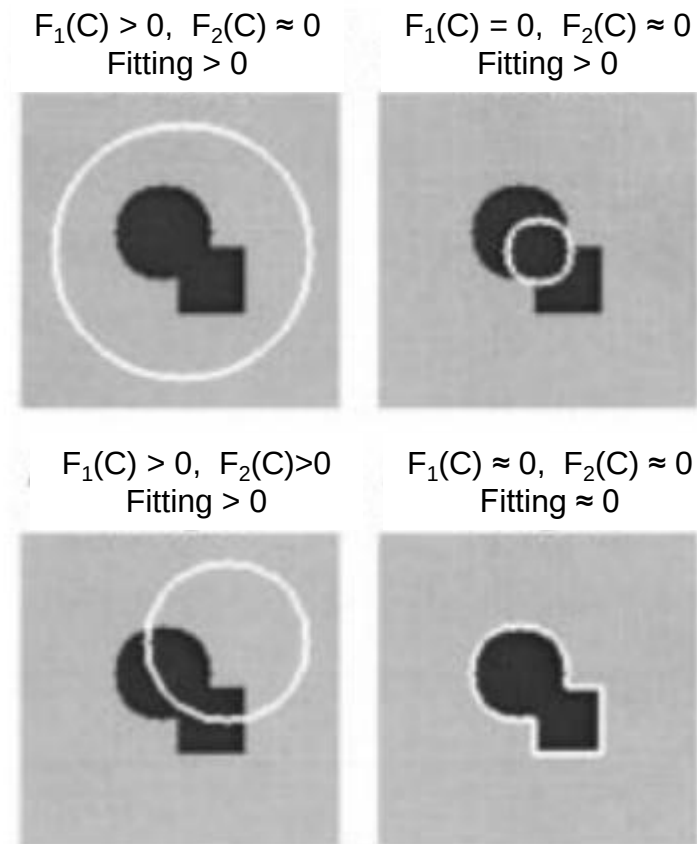
- Chan-Vese model: active contours without edges
- Limitation of edge-based level set
 - Cannot perform well if the edge is weak
 - The stopping function g is never zero on the edge
- Chan-Vese model defines a region-based energy function

$$g(I) = \frac{1}{1 + \|\nabla I\|^2}$$

类内方差最小化

$$F_1(C) + F_2(C) = \int_{\text{inside}(C)} |u_0(x, y) - c_1|^2 dx dy + \int_{\text{outside}(C)} |u_0(x, y) - c_2|^2 dx dy$$

C_1 and C_2 are average value of inside and outside regions, respectively.



Region-based Level set 基于区域的水平集

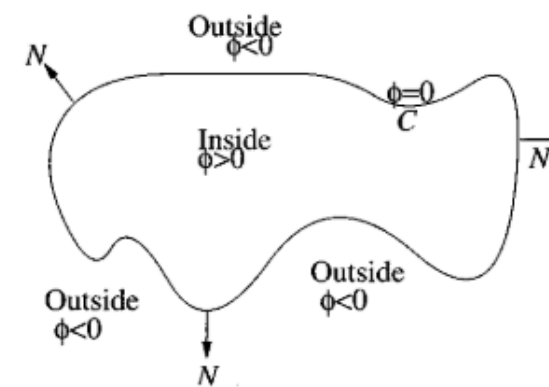


- Chan-Vese model: active contours without edges
- To add some regularizations, constraint the length and area of the curve

$$F(c_1, c_2, C) = \mu \cdot \text{Length}(C) + \nu \cdot \text{Area}(\text{inside}(C)) \\ + \lambda_1 \int_{\text{inside}(C)} |u_0(x, y) - c_1|^2 dx dy \\ + \lambda_2 \int_{\text{outside}(C)} |u_0(x, y) - c_2|^2 dx dy$$

Convert curve
representation
into level set
representation

$$F_\varepsilon(c_1, c_2, \phi) \\ = \mu \int_{\Omega} \delta_\varepsilon(\phi(x, y)) |\nabla \phi(x, y)| dx dy \quad \text{轮廓长度} \\ + \nu \int_{\Omega} H_\varepsilon(\phi(x, y)) dx dy \quad \text{区域面积} \\ + \lambda_1 \int_{\Omega} |u_0(x, y) - c_1|^2 H_\varepsilon(\phi(x, y)) dx dy \\ + \lambda_2 \int_{\Omega} |u_0(x, y) - c_2|^2 (1 - H_\varepsilon(\phi(x, y))) dx dy \quad \text{类内方差最小化}$$

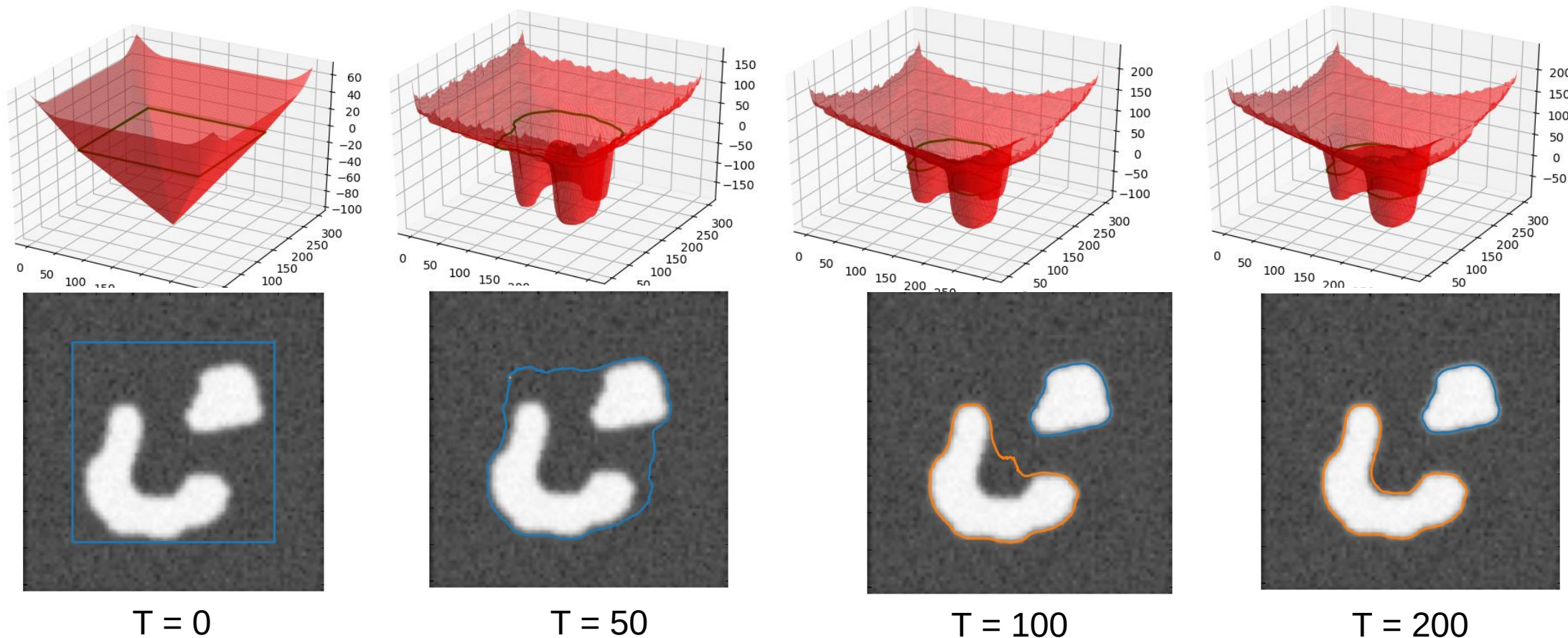


δ_ε : 脉冲函数

H_ε : 阶跃函数

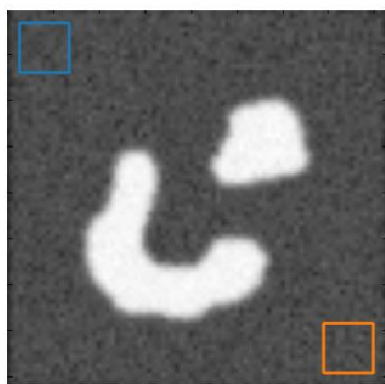
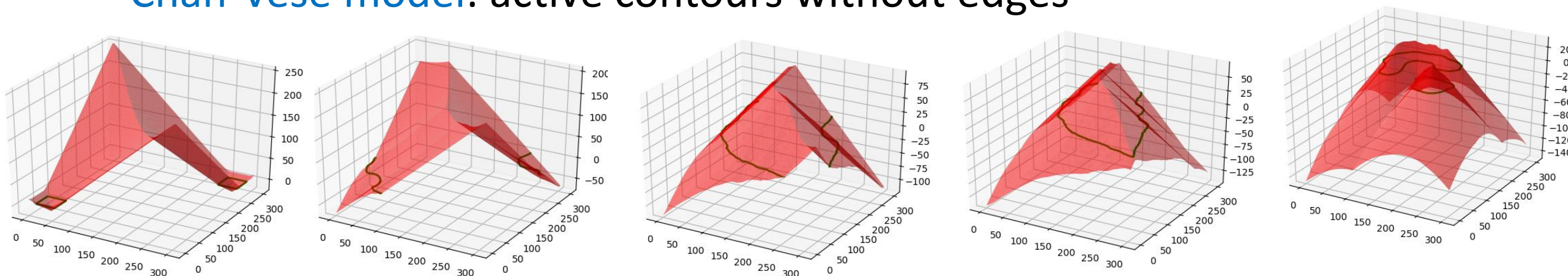
Region-based Level set 基于区域的水平集

- Chan-Vese model: active contours without edges

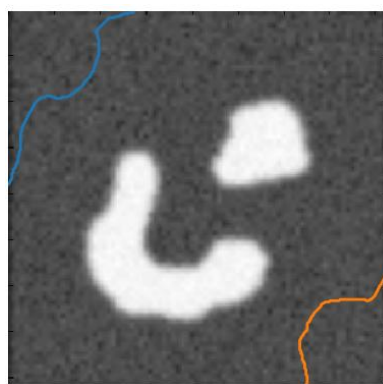


Region-based Level set 基于区域的水平集

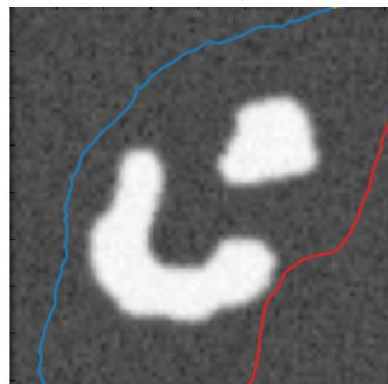
- Chan-Vese model: active contours without edges



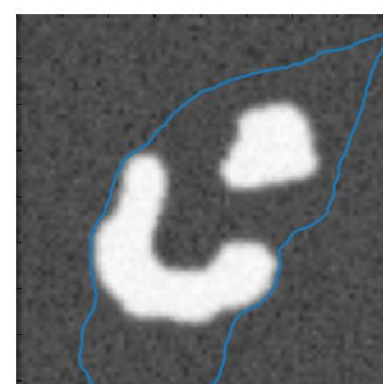
$T = 0$



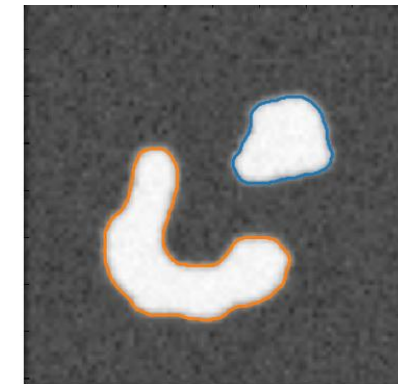
$T = 200$



$T = 500$



$T = 600$

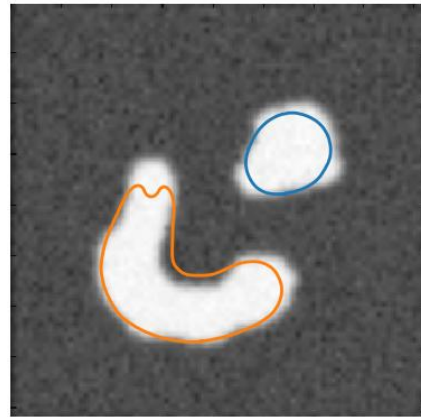
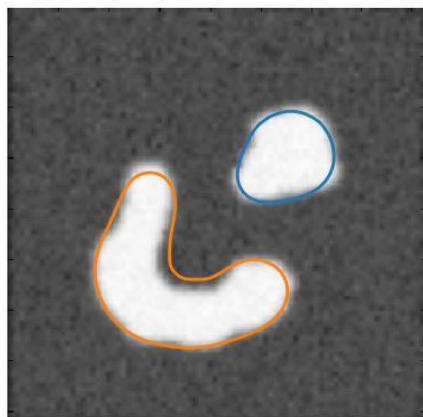
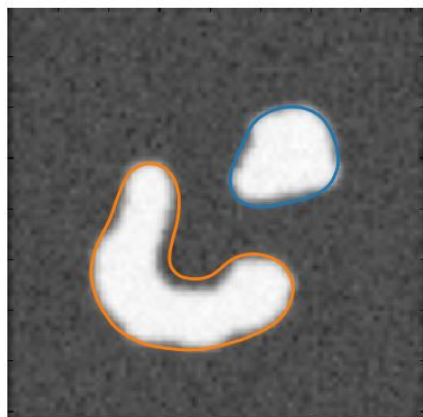


$T = 800$

Region-based Level set 基于区域的水平集



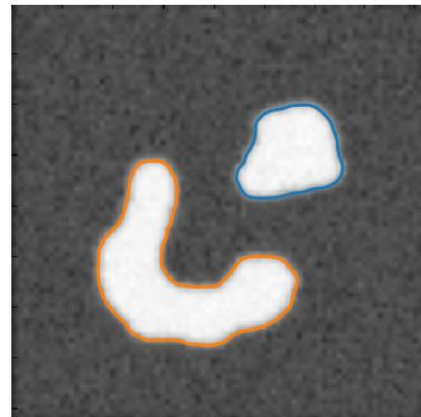
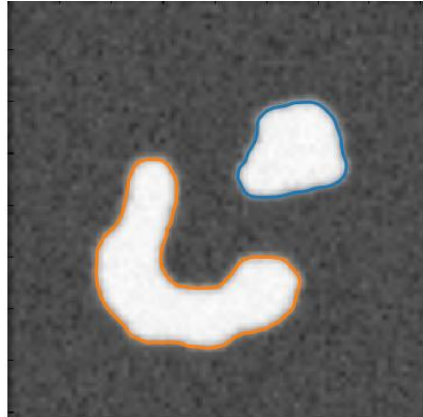
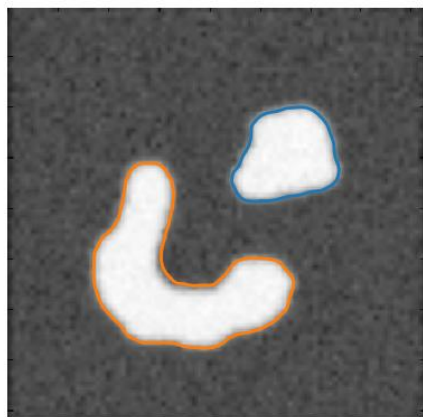
- Comparison between edge-based and Chan-Vese model



Edge-based level set

May pass through the boundary due to non-zero stopping function

$$g(I) = \frac{1}{1 + \|\nabla I\|^2}$$



Chan-Vese model

No need edge detection
Better fit the boundary

T = 200

T = 300

T = 400

Outline



1, Intensity-based Segmentation

- Thresholding

- Watershed

2, Clustering-based Segmentation

3, Active contours

- Snake

- Level set

4, Graph-based segmentation

5, Post-processing (morphological operation)

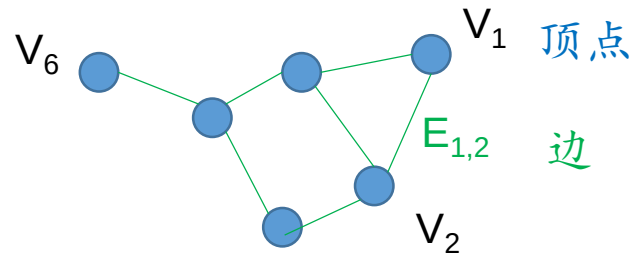
Graph-based segmentation 基于图的分割



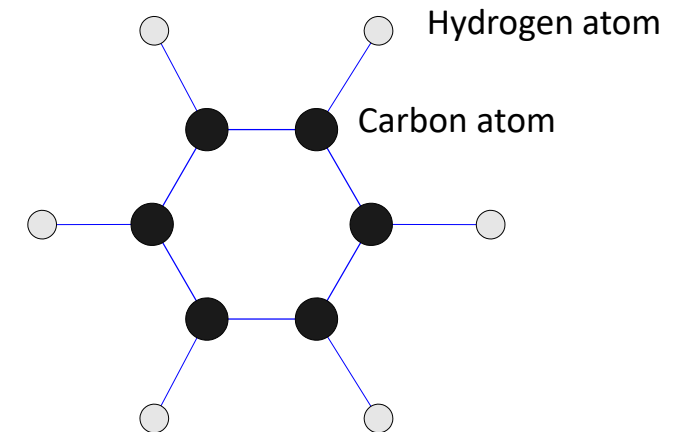
What is a graph?

A graph is a set of **vertices** linked by some **edges**, used to represent:

- Relationship between different objects
- Links between components of an object



Find a path in the map

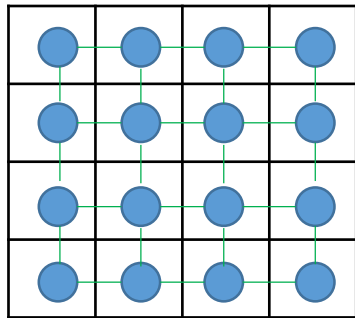


Benzene Molecule: C₆H₆

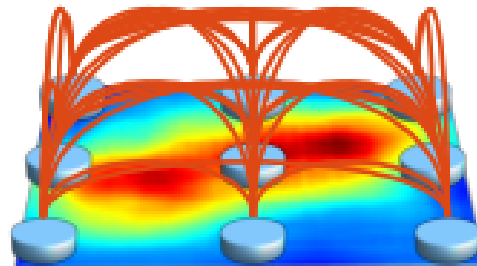
Graph-based segmentation 基于图的分割

An image can also be seen as a graph

- Each pixel is a vertex
- How to define the edge?

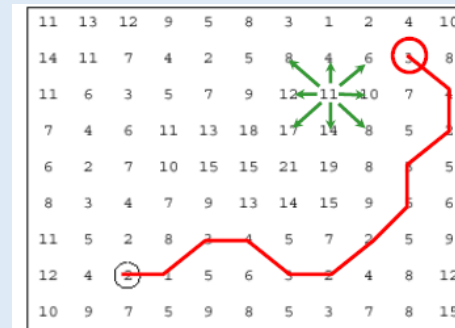


Neighborhood connection



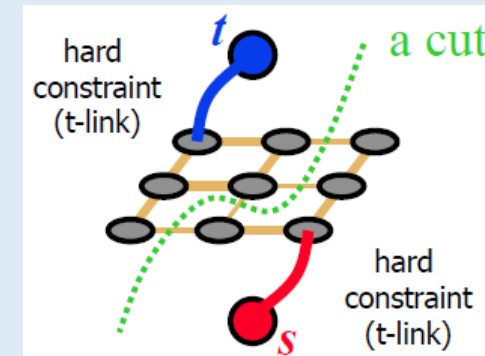
Fully connected

Graph-based Segmentation



最短路径

Shortest paths
approach



图割

Graph cuts
approach

Graph-based segmentation 基于图的分割



Intelligent Scissors 智能剪刀

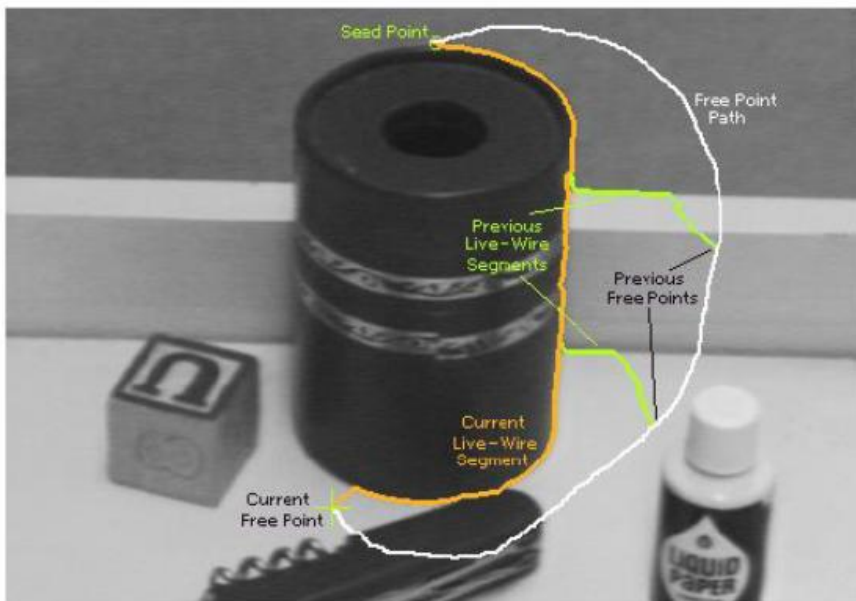


Figure 2: Image demonstrating how the live-wire segment adapts and snaps to an object boundary as the free point moves (via cursor movement). The path of the free point is shown in white. Live-wire segments from previous free point positions (t_0 , t_1 , and t_2) are shown in green.

[Eric Mortensen, William Barrett, 1995]



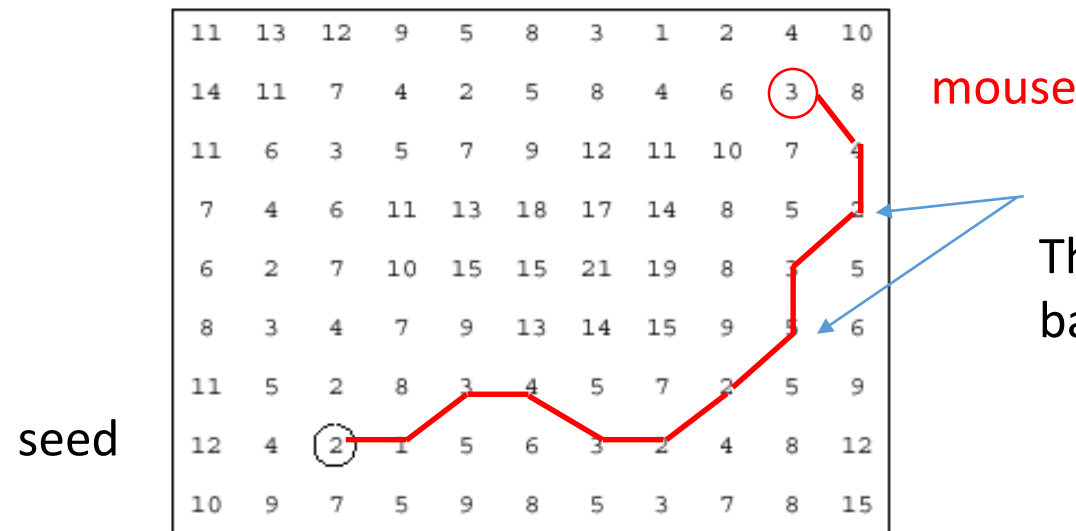
Photoshop的磁性套索工具

Graph-based segmentation 基于图的分割



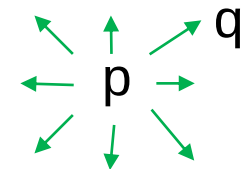
Intelligent Scissors 智能剪刀

- find lowest cost path from seed to mouse on a graph (e.g. N8 pixel grid)
- weighted by intensity contrast



$$c_p = \frac{25}{1 + |\nabla I_p|}$$

The cost can be defined based on the local contrast



The weight between p and q can be defined as $w_{pq} = c_q$

Use node-weighted version of “shortest paths” (Dijkstra) 沿着边缘点附近的最短路径搜索



The path with the lowest cost

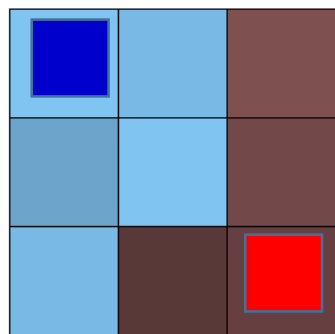
Graph-based segmentation 基于图的分割



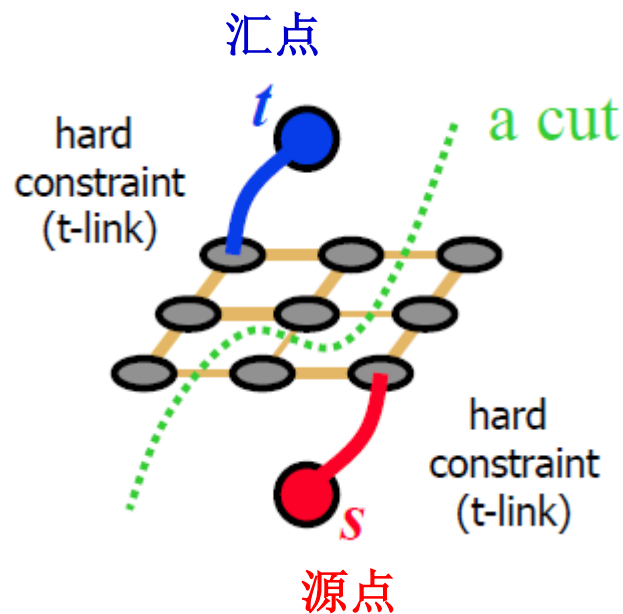
Graph Cut 图割

一个“割” (cut) 是图中一部分边的集合，将图分成两个不相连的部分，即得到分割结果

背景种子点



前景种子点



额外增加两个点：源点s和汇点t,分别代表两个类别

边分为两种：**n-link**: 相邻两个像素间的权重

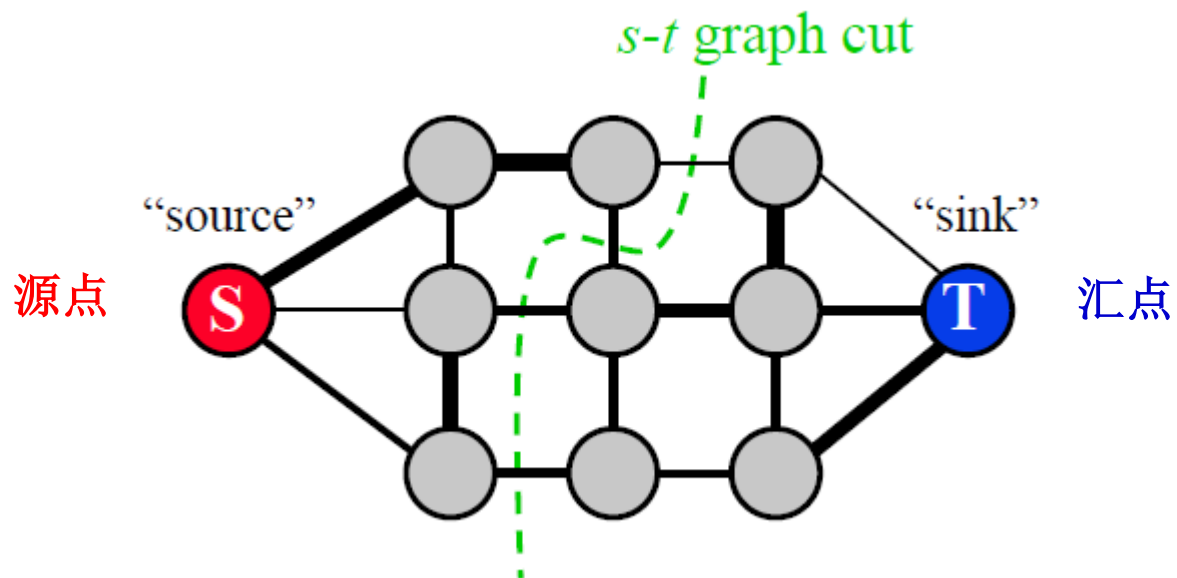
t-link: 某个像素与s/t之间的连接权重

Graph-based segmentation 基于图的分割



Graph Cut

- **Goal:** divide the graph into two parts separating **source** and **sink** nodes



最小割 (Min-cut): 代价 (权重) 之和最小的一个割, 作为分割的边界

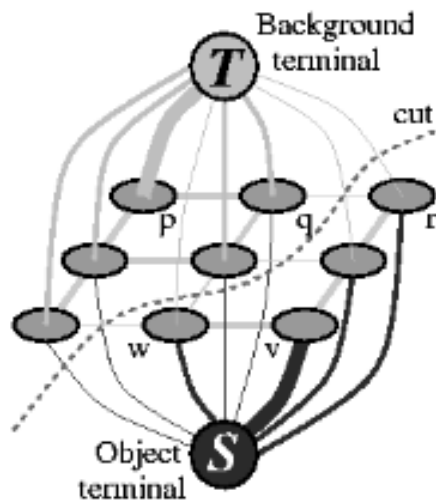
- **Min cut problem:** Delete “best” set of edges to disconnect t from s. **最小割**
- This is also called a **max-flow** problem **最大流**

Graph-based segmentation 基于图的分割



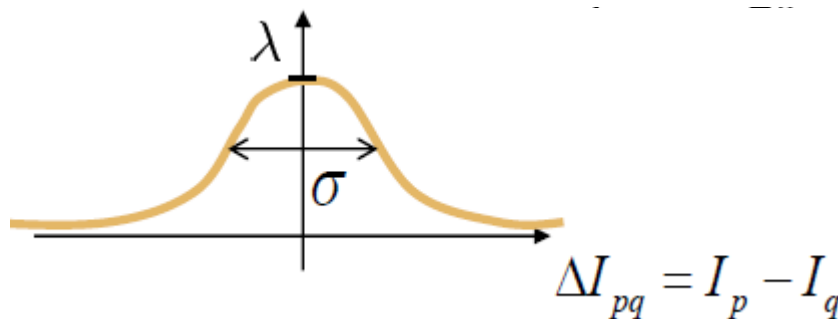
Graph Cut

如何设置图中边的权重（代价）？



n-link: 相邻两个像素间的权重
一般相邻像素间的灰度差异越大，代价越小

$$B_{\{p,q\}} \propto \exp\left(-\frac{(I_p - I_q)^2}{2\sigma^2}\right) \cdot \frac{1}{\text{dist}(p,q)}$$

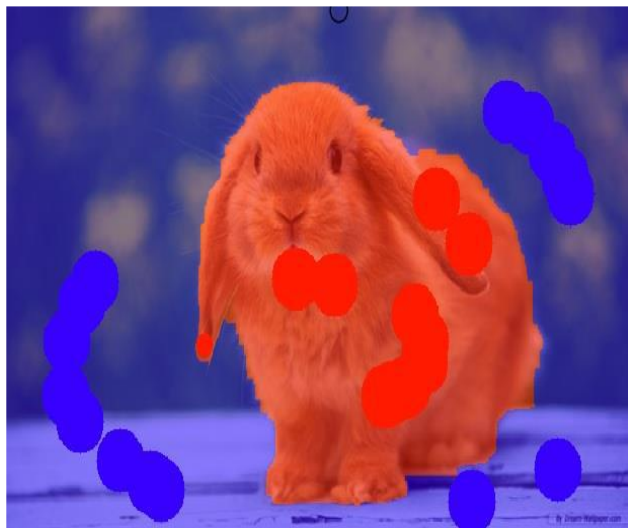


Graph-based segmentation 基于图的分割

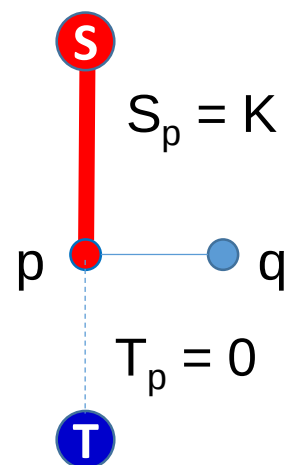


Graph Cut

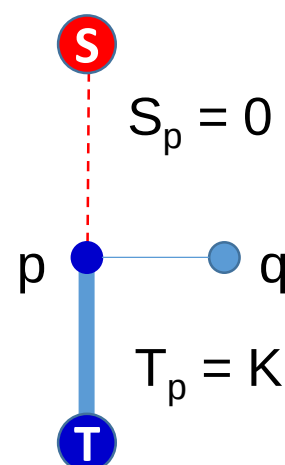
如何设置图中边的权重（代价）？ **t-link**: 某个像素与s或t之间的连接权重, 表示为 S_p 或 T_p



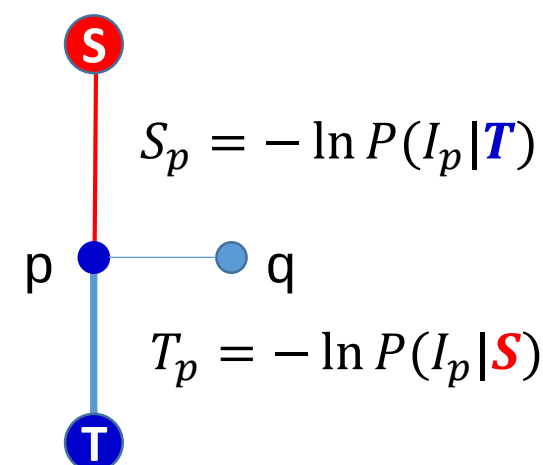
若 p 被指定为前景



若 p 被指定为背景



其他

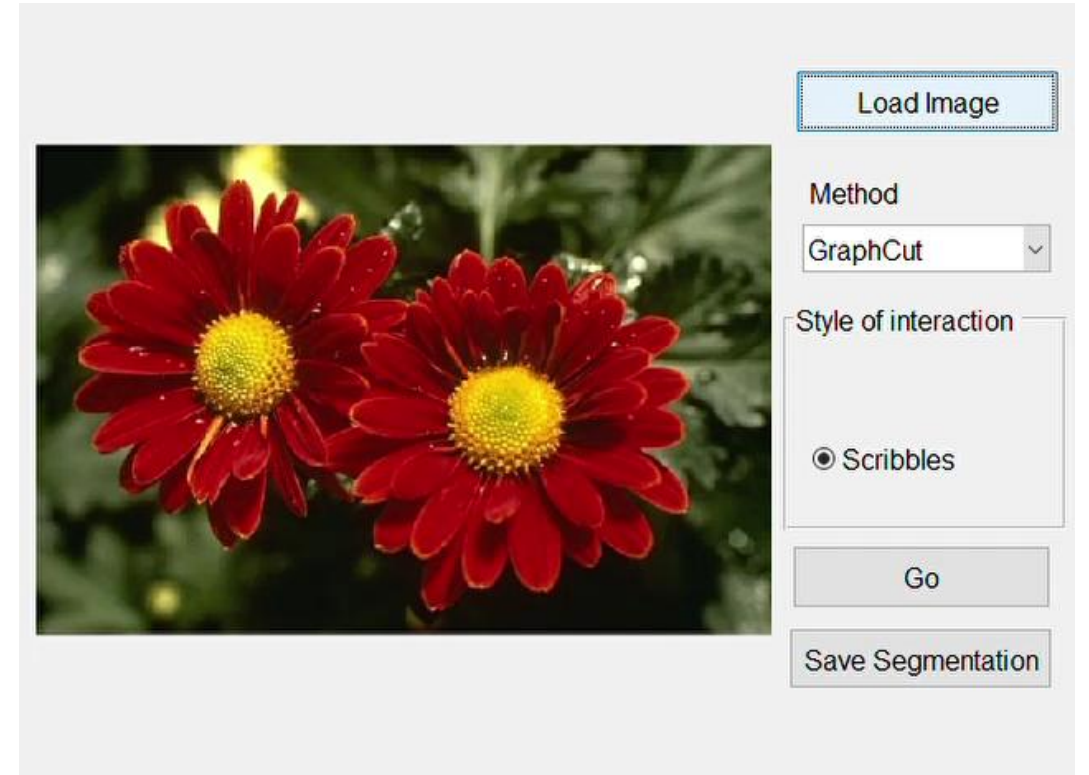
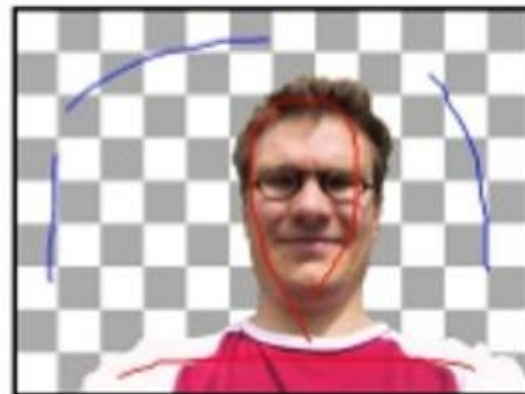


$$K = \max(B\{p, q\}) + 1$$

Graph-based segmentation 基于图的分割



Graph Cut Results



Graph-based segmentation 基于图的分割



交互式分割（抠图）



Outline



1, Intensity-based Segmentation

- Thresholding

- Watershed

2, Clustering-based Segmentation

3, Active contours

- Snake

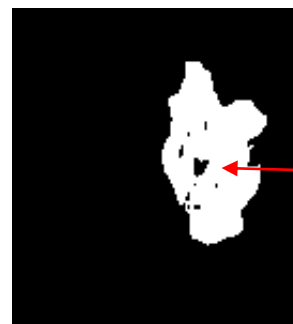
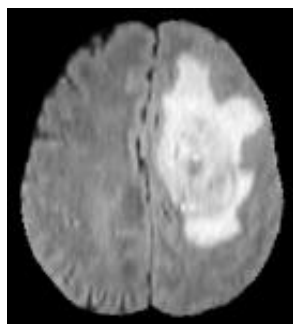
- Level set

4, Graph-based segmentation

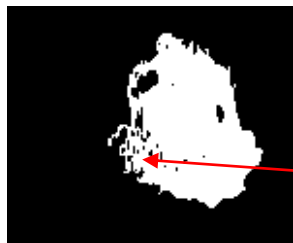
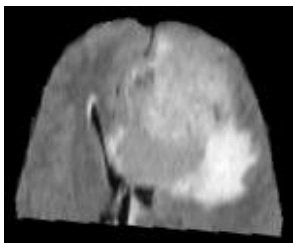
5, Post-processing (morphological operation)

Why post-processing ?

- The initial automatic segmentation are often **noisy**



Small holes



Rough
boundary

Image

Ground truth

Segmentation

morphological operation 形态学运算

morphological operation

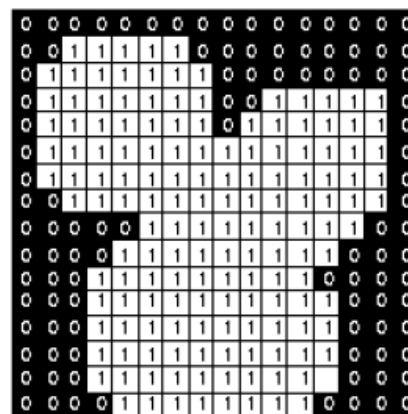
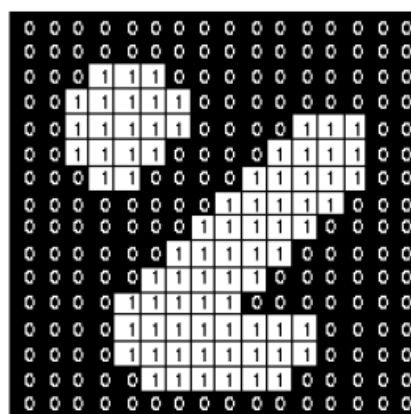
- A simple and widely used for post-processing
- So that the segmentation is less noisy and more smooth
- It can also be used to extract centerline of an object
- Basic operations include: dilation, erosion, opening, closing

膨胀

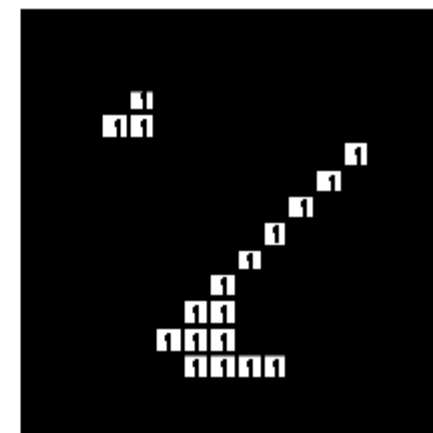
腐蚀

开运算

闭运算



Dilation 膨胀



Erosion 腐蚀

morphological operation 形态学运算



- **Value** of each pixel in the output image is based on a **comparison** of the corresponding **pixel** in the input image with a **structure element**

腐蚀：逻辑“与”操作

0	1	1	1	0
0	1	1	1	0
0	1	1	1	0
0	1	1	1	1
0	1	1	1	0

image

1	1	1
1	1	1
1	1	1

Structure element
结构单元

例如：

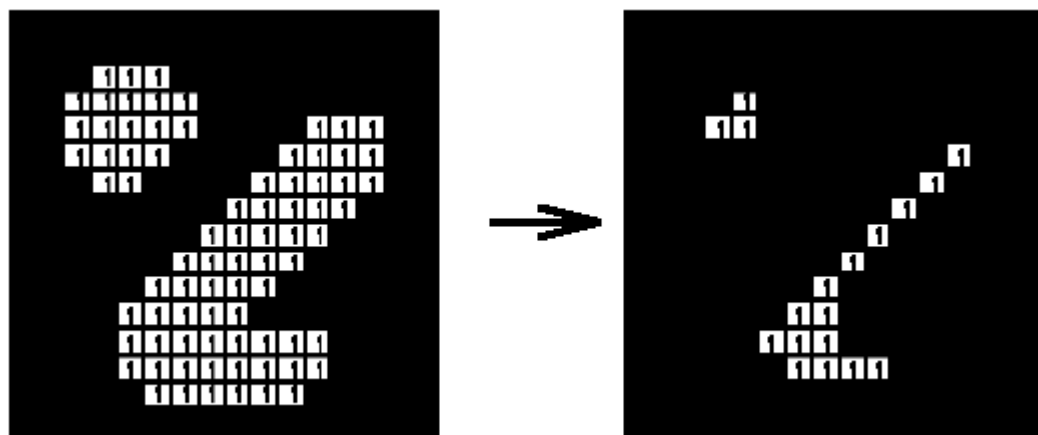
若图像的局部范围内与结构单元的值全部一样，输出为1，否则为0

0	1	1	1	0
0	1	1	1	0
0	1	1	1	0
0	1	1	0	1
0	1	1	1	0

output

morphological operation 形态学运算

- A first example: erosion 腐蚀：逻辑“与”操作
 - Erosion is an important morphological operation



若图像的局部范围内与结构单元的值全部一样，输出为1，否则为0

Structure element

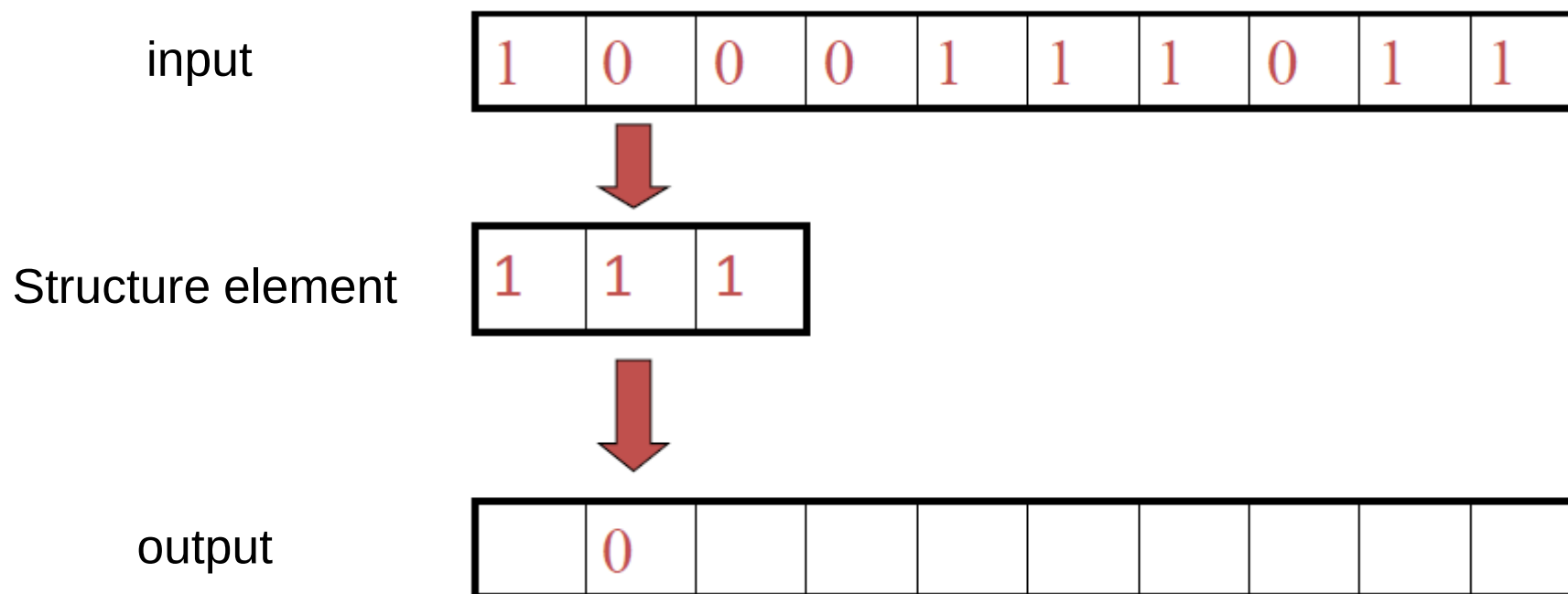
1	1	1
1	1	1
1	1	1

If the local 3x3 patch has the same pattern as the structure element foreground, the output value in the center is 1, otherwise 0

morphological operation 形态学运算



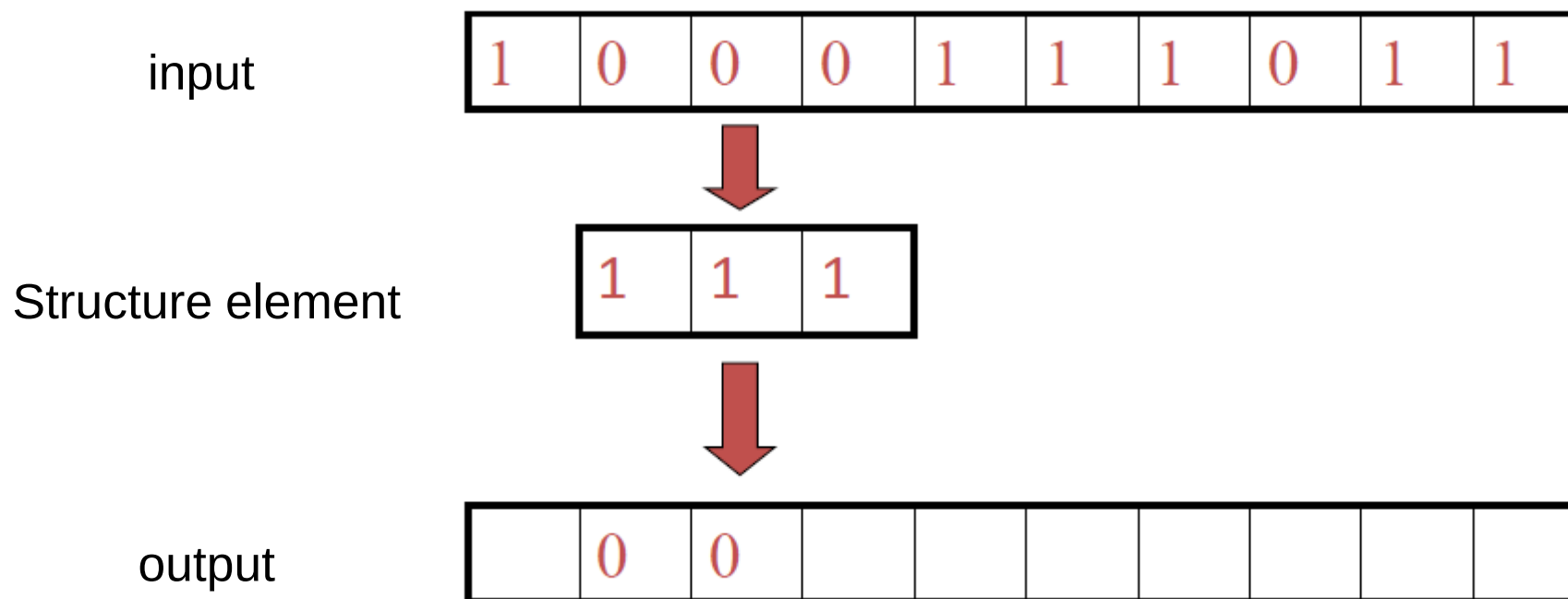
- A first example: erosion 腐蚀：逻辑“与”操作



morphological operation 形态学运算



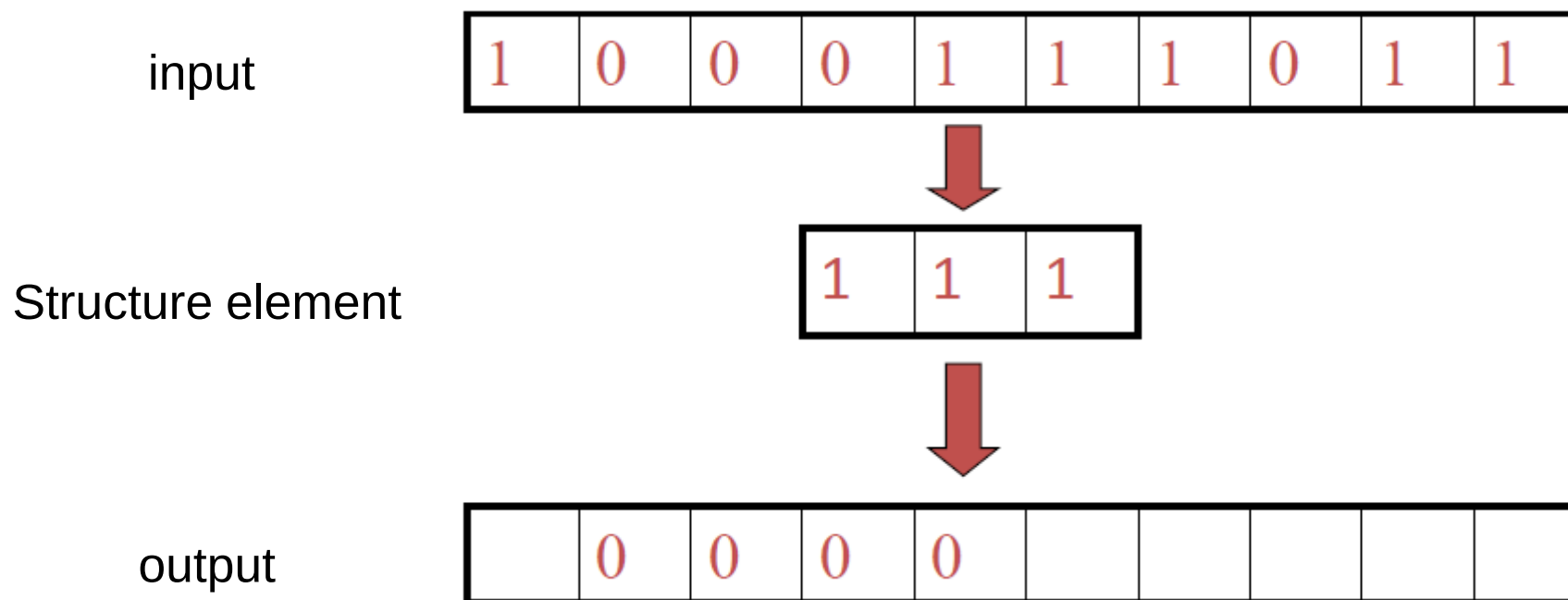
- A first example: erosion 腐蚀：逻辑“与”操作



morphological operation 形态学运算



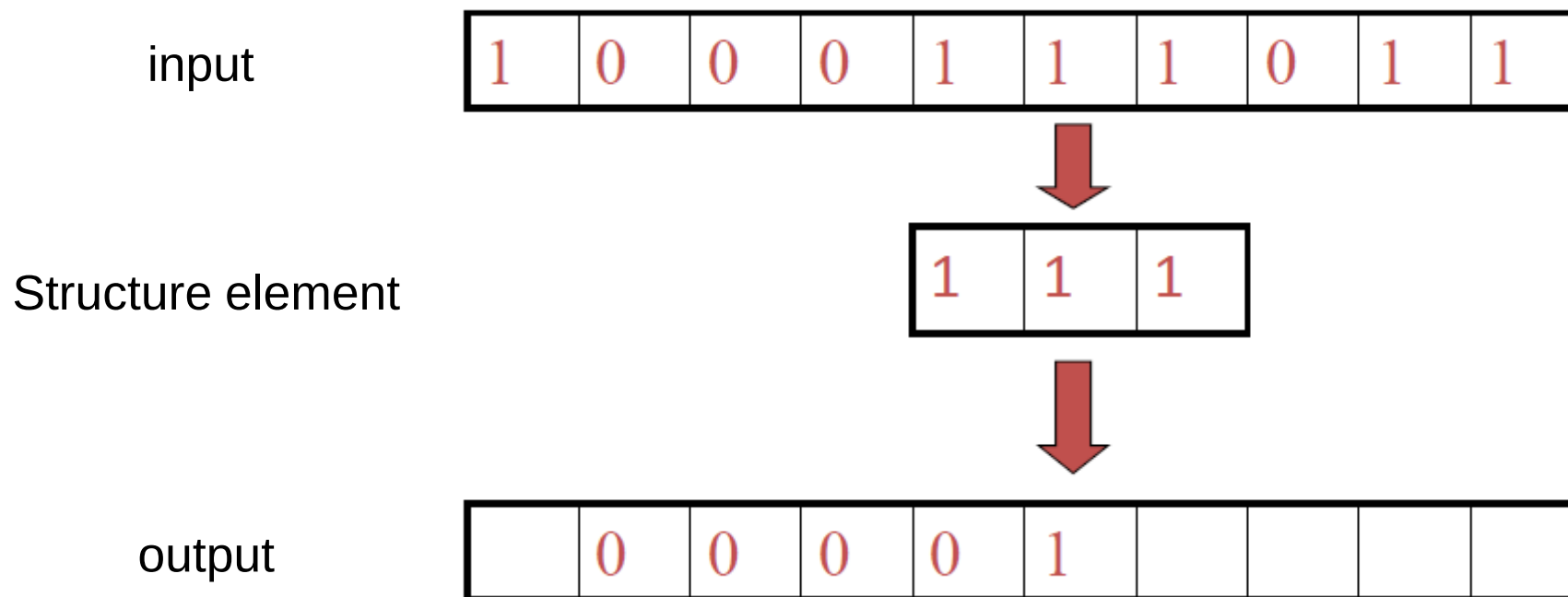
- A first example: erosion 腐蚀：逻辑“与”操作



morphological operation 形态学运算



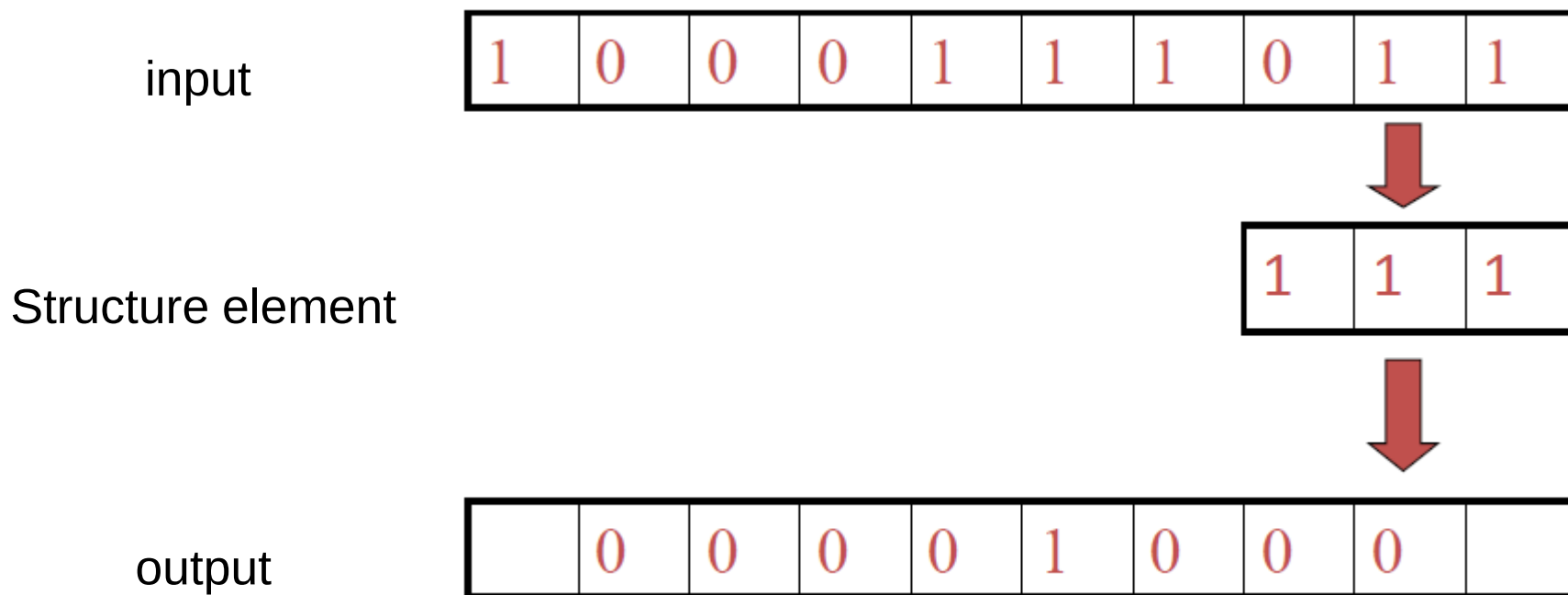
- A first example: erosion 腐蚀：逻辑“与”操作



morphological operation 形态学运算



- A first example: erosion 腐蚀：逻辑“与”操作

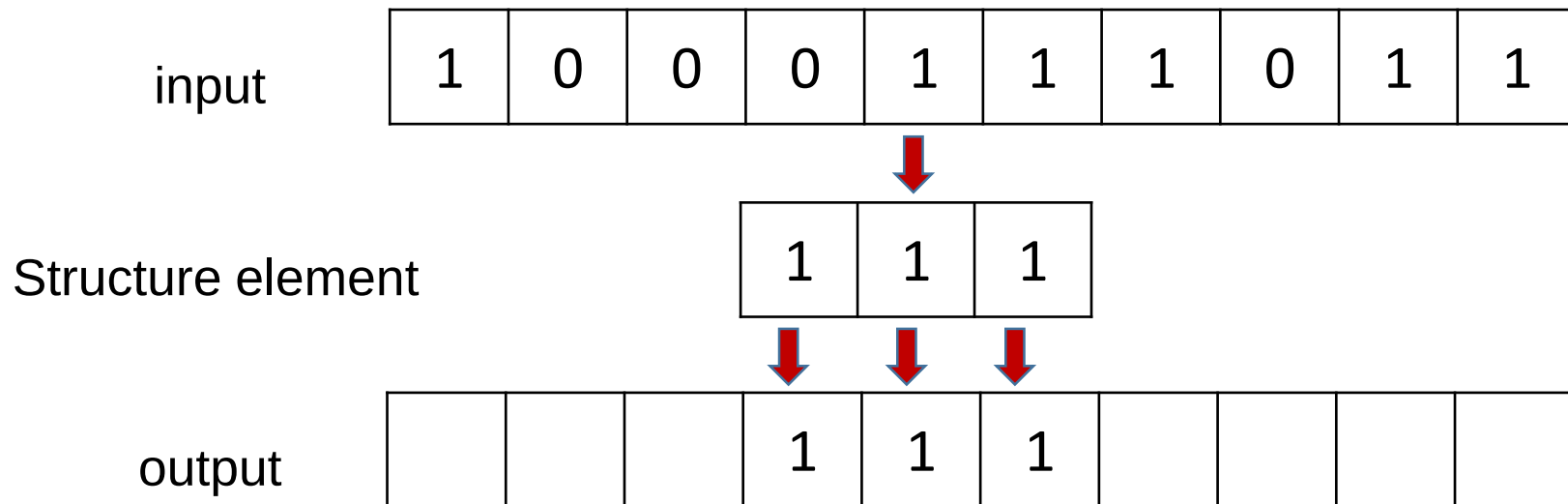


腐蚀可去掉图像中小的不联通区域

morphological operation 形态学运算



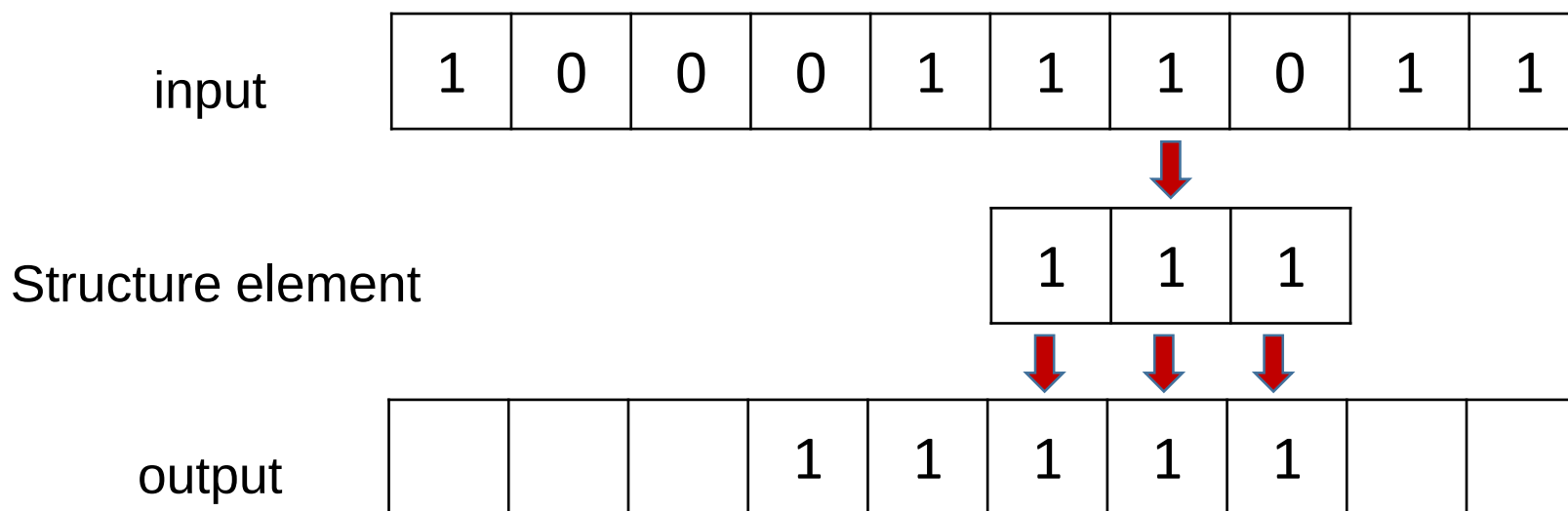
- Dilation 膨胀：逻辑“或”操作
 - Expand the foreground region with the structure element
 - Put the structure element **at each foreground pixel**
 - The neighboring is set based on an “OR” operation



morphological operation 形态学运算



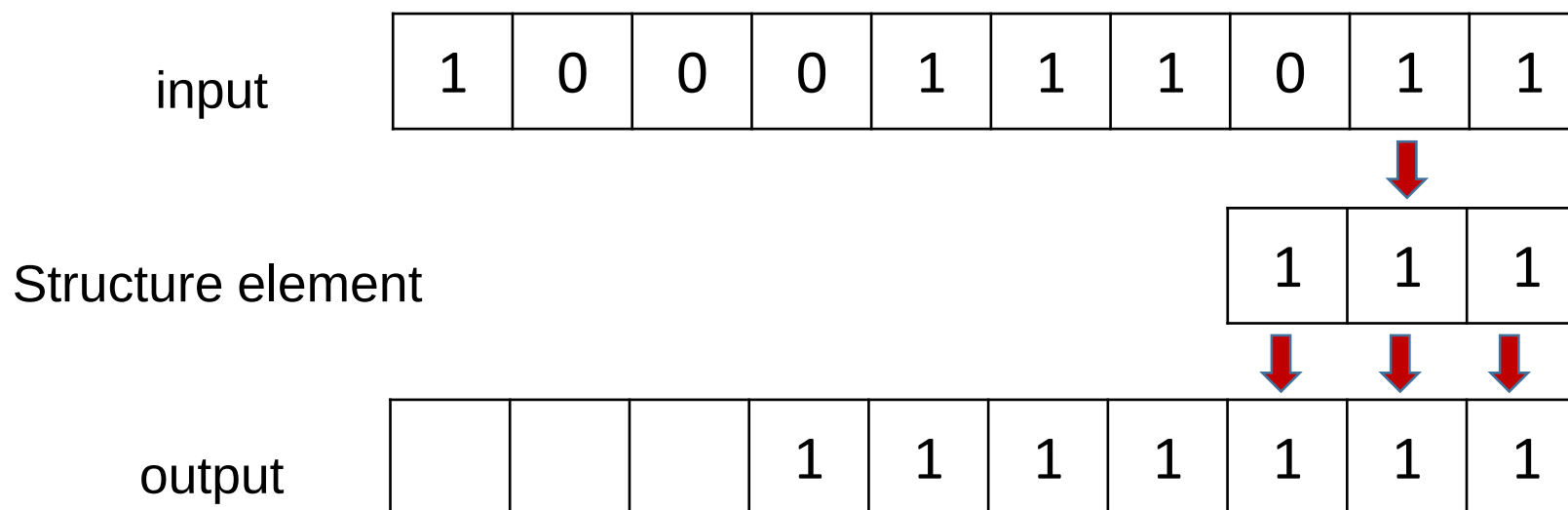
- Dilation 膨胀：逻辑“或”操作
 - Expand the foreground region with the structure element
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morphological operation 形态学运算



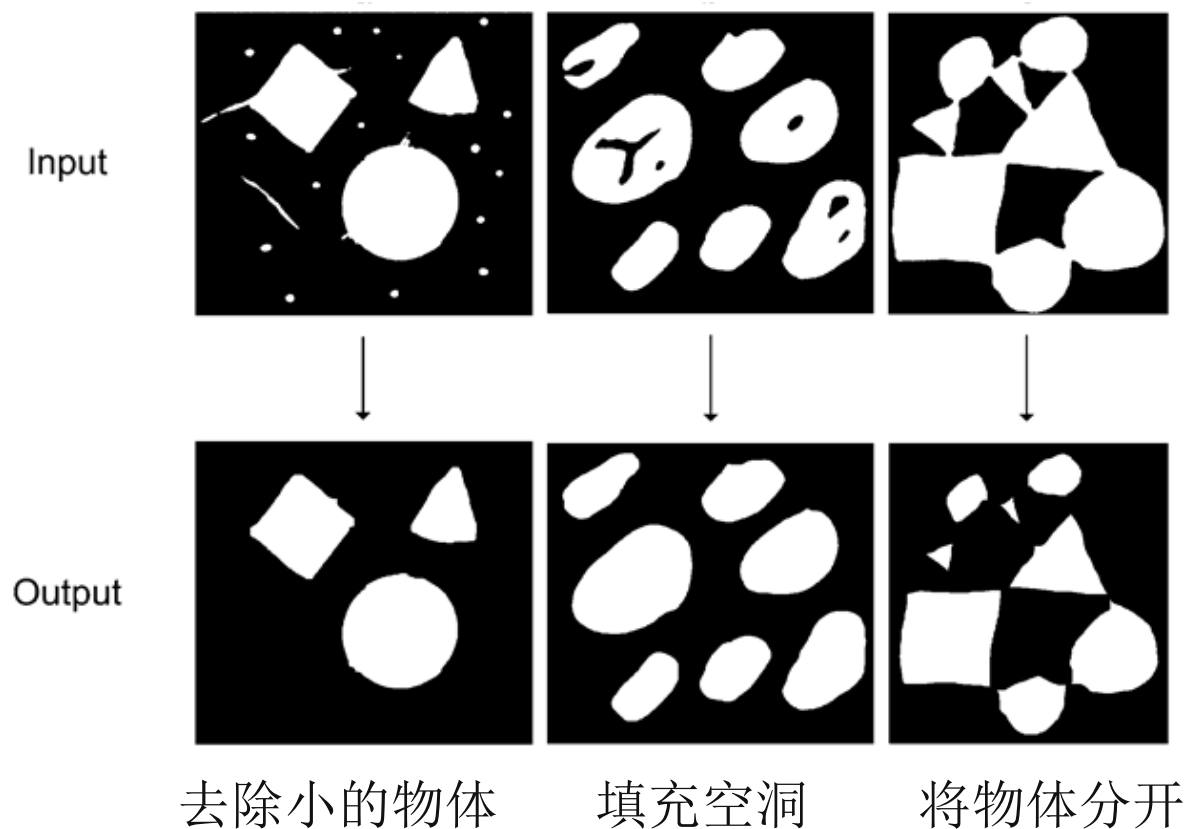
- Dilation 膨胀：逻辑“或”操作
 - Expand the foreground region with the structure element
 - Put the structure element at **each foreground pixel**
 - The neighboring is set based on an “OR” operation



膨胀可填充图像中小的空洞

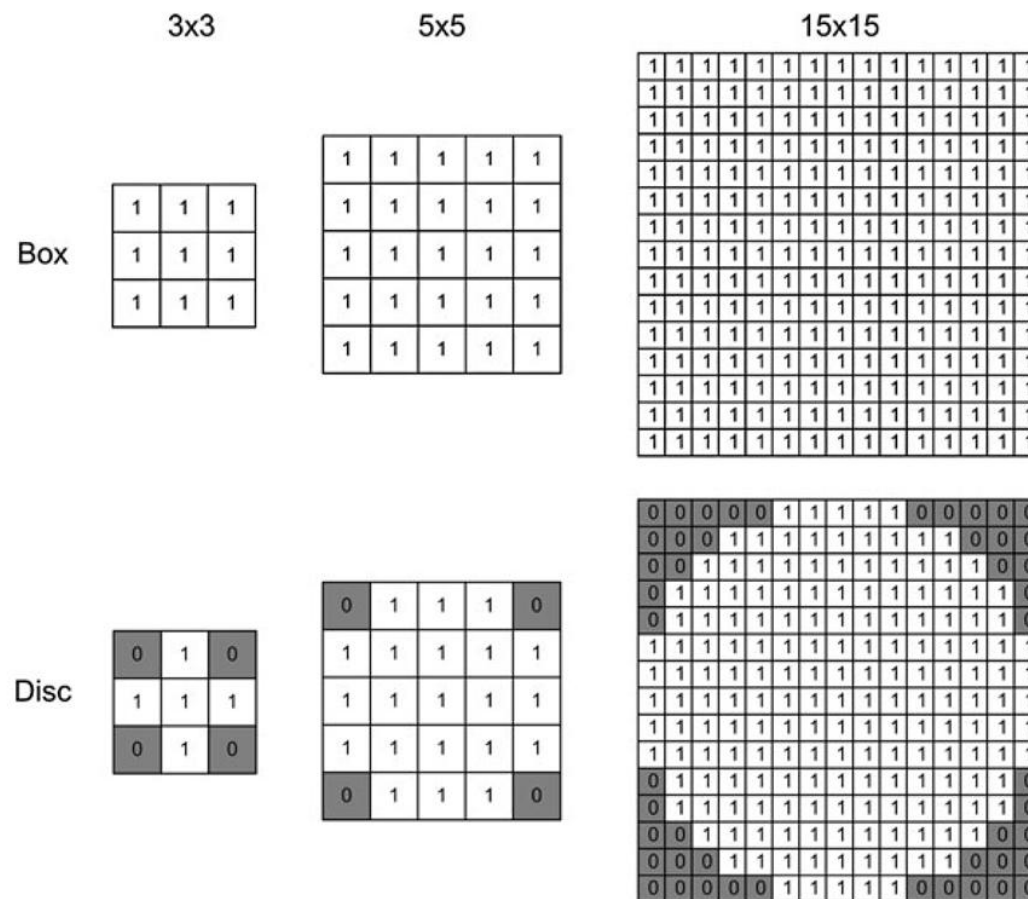
morphological operation 形态学运算

- Effect of dilation and erosion



morphological operation 形态学运算

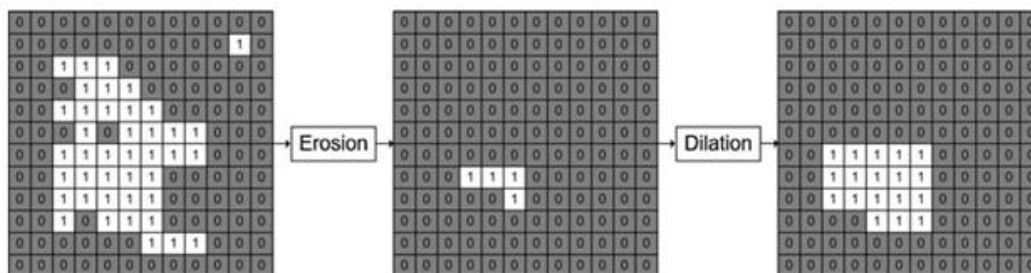
- The structure element can be set with different sizes



morphological operation 形态学运算

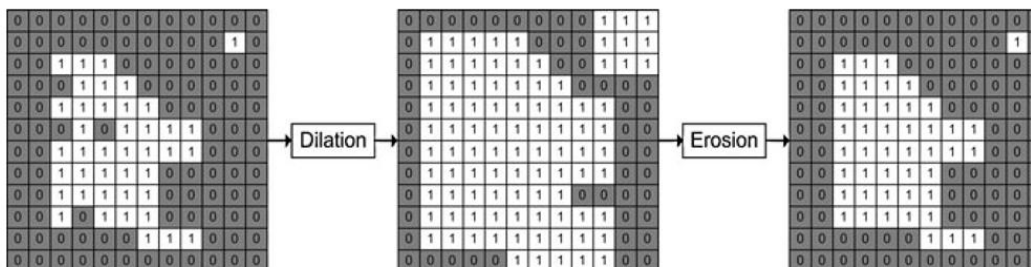
- Opening and closing 开运算与闭运算

- Opening:** An erosion followed by dilation 先腐蚀后膨胀



Opening tends to “open” **small gaps** or spaces between touching objects in an image.

- Closing:** A dilation followed by erosion 先膨胀后腐蚀

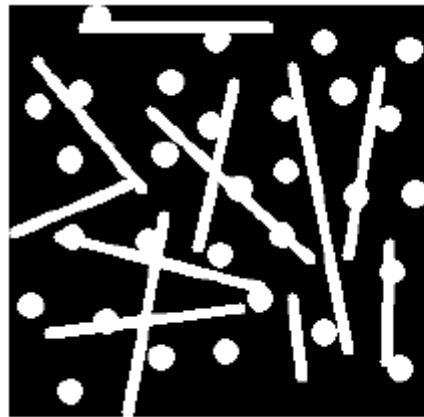


Closing tends to “close” **small holes** or spaces in an image.

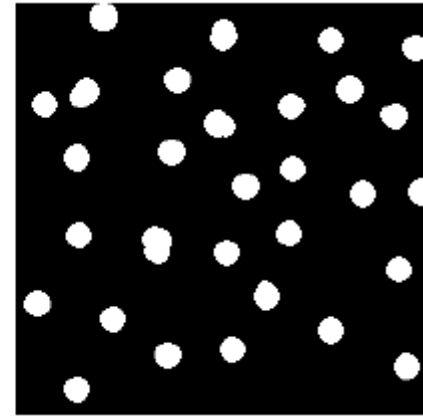
morphological operation 形态学运算



- Opening 开运算 (先腐蚀后膨胀)
 - Separate out the circles from the lines



A mixture of circle and lines



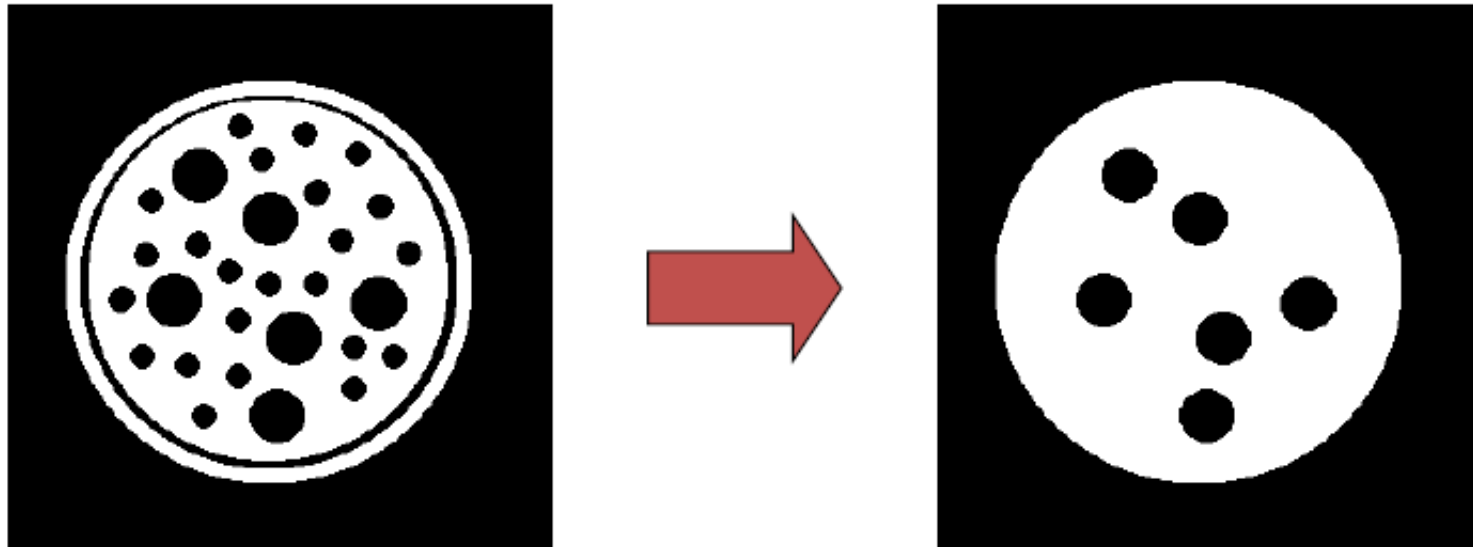
Opening with a disk shaped structuring element with 11 pixels in diameter

The lines have been almost completely removed while the circles remain almost completely unaffected.

morphological operation 形态学运算



- Closing 闭运算（先膨胀后腐蚀）
 - Closing operation with a 22 pixel disc
 - Closes small holes in the foreground



小结

- 水平集
 - 基于边缘的方法
 - 基于区域的方法
- 基于图的分割
 - 智能剪刀：最短路径
 - 图割：最小割
- 形态学运算
 - 膨胀、腐蚀
 - 开运算、闭运算

$$\frac{\partial \phi}{\partial t} = F |\nabla \phi|$$

